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with MON [Table 1] and summary of secondary outcome—AEs which could not be performed meta-analysis comparing SFC with MFC [Table 2]. AEs: Adverse events; MFC: Montelukast/fluticasone; MON: Montelukast; SFC: Salmeterol/fluticasone.

SI Appendix 9: This section reports the results of meta-analysis forest plots of AEs after 12 weeks treatment comparing SFC with MFC. AEs: Adverse events; MFC: Montelukast/fluticasone; SFC: Salmeterol/flu

Appendix 1: Search strategy

A. Cochrane Library

#1 MeSH descriptor: [Fluticasone] explode all trees 1581

#2 MeSH descriptor: [Salmeterol Xinafoate] explode all trees 1024

#3 #1 and #2 551

#4 MeSH descriptor: [Fluticasone-Salmeterol Drug Combination] explode all trees 180

#5 (Seretide or Advair or Atmadisc or Viani or (fluticasone and salmeterol)):ti,ab,kw (Word variations have been searched) 2030

#6 #3 or #4 or #5 2030

#7 MeSH descriptor: [Budesonide] explode all trees1708

#8 MeSH descriptor: [Formoterol Fumarate] explode all trees 981

#9 #7 and #8 377

#10 MeSH descriptor: [Budesonide, Formoterol Fumarate Drug Combination] explode all trees 149

#11 (Symbicort or rilast or "duoresp spiromax" or "bufomix easyhaler" or (budesonide and formoterol)):ti,ab,kw (Word variations have been searched) 1362

#12 (fostair or (beclometasone and formoterol and glycopyrronium)):ti,ab,kw (Word variations have been searched) 52

#13 (singulair or montelukast):ti,ab,kw (Word variations have been searched) 1707

#14 #9 or #10 or #11 or #12 or #13 3082

#15 #6 and #14 359

#16 MeSH descriptor: [Asthma] explode all trees 11093

#17 MeSH descriptor: [Respiratory Sounds] explode all trees 602

#18 (asthma* or "lung allergy" or "lung allergies"):ti,ab,kw (Word variations have been searched) 32931

#19 (antiasthma* or anti-asthma*):ti,ab,kw (Word variations have been searched) 2045

#20 #16 or #17 or #18 or #19 33360

#21 #15 and #20 277

#22 (child* or teen* or adolescen* or juvenile or "young person" or "young people" or minors or youth*):ti,ab,kw (Word variations have been searched) 252612

#23 MeSH descriptor: [Child] explode all trees 1203

#24 MeSH descriptor: [Adolescent] explode all trees 101821

#25 MeSH descriptor: [Minors] explode all trees 8

#26 #22 or #23 or #24 or #25 252612

#27 #21 and #26 108

#28 #21 and #26 in Trials 104

B. MEDLINE

1 exp Fluticasone/ and exp Salmeterol Xinafoate/ (926)

2 "Fluticasone-Salmeterol Drug Combination"/ (390)

3 (Seretide or Advair or Atmadisc or Viani or (fluticasone and salmeterol)).tw,kw,kf. (1162)

4 or/1-3 (1371)

5 exp Budesonide/ and exp Formoterol Fumarate/ (617)

6 "Budesonide, Formoterol Fumarate Drug Combination"/ (146)

7 (Symbicort or rilast or "duoresp spiromax" or "bufomix easyhaler" or (budesonide and formoterol)).tw,kw,kf. (815)

8 (fostair or (beclometasone and formoterol and glycopyrronium)).tw,kw,kf. (18)

9 (singulair or montelukast).tw,kw,kf. (2139)

10 or/5-9 (3044)

11 4 and 10 (342)

12 exp ASTHMA/ (125202)

13 Respiratory Sounds/ (8726)

14 (asthma* or "lung allerg*").tw,kw,kf. (154887)

15 (antiasthma* or anti-asthma*).tw,kw,kf. (2955)

16 or/12-15 (179165)

17 (Child* or teen* or adolescen* or juvenile or "young person" or "young people" or minors or youth*).tw,kw,kf. (1644241)

18 exp CHILD/ (1865363)

19 exp CHILD, PRESCHOOL/ (896883)

20 exp ADOLESCENT/ (1975162)

21 exp MINORS/ (2547)

22 or/17-21 (3454123)

23 11 and 16 and 22 (106)

24 exp clinical trial/ (844779)

25 exp randomized controlled trials/ (131622)

26 exp double-blind method/ (154931)

27 exp single-blind method/ (27714)

28 exp cross-over studies/ (46596)

29 randomized controlled trial.pt. (495639)

30 clinical trial.pt. (519829)

31 controlled clinical trial.pt. (93449)

32 (clinic* adj2 trial).mp. (706153)

33 (random* adj5 control* adj5 trial*).mp. (705593)

34 (crossover or cross-over).mp. (93640)

35 ((singl* or double* or trebl* or tripl*) adj (blind* or mask*)).mp. (230948)

36 randomi*.mp. (858395)

37 (random* adj5 (assign* or allocat* or assort*)).mp. (237695)

38 or/24-37 (1391725)

39 23 and 38 (69)

40 animals/ not humans/ (4615830)

41 39 not 40 (69)

C. EMBASE

1 exp fluticasone/ and exp salmeterol/ (2393)

2 exp fluticasone propionate plus salmeterol/ (3494)

3 (Seretide or Advair or Atmadisc or Viani or (fluticasone and salmeterol)).tw,kw. (3222)

4 or/1-3 (6339)

5 exp budesonide/ and exp formoterol/ (2218)

6 exp budesonide plus formoterol/ (2417)

7 (Symbicort or rilast or "duoresp spiromax" or "bufomix easyhaler" or (budesonide and formoterol)).tw,kw. (2125)

8 (fostair or (beclometasone and formoterol and glycopyrronium)).tw,kw. (104)

9 (singulair or montelukast).tw,kw. (4523)

10 or/5-9 (8771)

11 4 and 10 (2292)

12 exp asthma/ (249982)

13 exp abnormal respiratory sound/ (56812)

14 (asthma* or "lung allerg*").tw,kw. (222419)

15 (antiasthma* or anti-asthma*).tw,kw. (4239)

16 or/12-15 (323913)

17 (Child* or teen* or adolescen* or juvenile or "young person" or "young people" or minors or youth*).tw,kw. (1997179)

18 exp juvenile/ (3315040)

19 exp child/ (2525900)

20 exp adolescent/ (1473087)

21 or/17-20 (3831382)

22 11 and 16 and 21 (437)

23 (clin* adj2 trial).mp. (1523054)

24 ((singl* or doubl* or trebl* or tripl*) adj (blind* or mask*)).mp. (299059)

25 (random* adj5 (assign* or allocat*)).mp. (186392)

26 randomi*.mp. (1184992)

27 crossover.mp. (96132)

28 exp randomized-controlled-trial/ (584064)

29 exp double-blind-procedure/ (168295)

30 exp crossover-procedure/ (61602)

31 exp single-blind-procedure/ (37435)

32 exp randomization/ (85394)

33 or/23-32 (2234934)

34 22 and 33 (211)

35 exp Animals/ or exp Invertebrate/ or Animal Experiment/ or Animal Model/ or Animal Tissue/ or Animal Cell/ or Nonhuman/ (26707201)

36 Human/ or Normal Human/ or Human Cell/ (20419345)

37 35 not 36 (6350196)

38 34 not 37 (211)

D. CNKI（期刊学位会议）

(SU % '沙美特罗替卡松' OR SU=(舒利迭+沙美特罗*氟替卡松+Seretide+Advair)) AND SU=(哮喘)*(儿童+青少年+学龄前+学龄+适龄+幼儿+少儿+患儿)*(临床试验+临床研究+临床评估+临床评价+效果评价+干预研究+疗效+随机+盲法+双盲+单盲+对照+三盲+队列+RCT) OR (AB % '沙美特罗替卡松' OR AB=(舒利迭+沙美特罗*氟替卡松+Seretide+Advair)) AND AB=(哮喘)*(儿童+青少年+学龄前+学龄+适龄+幼儿+少儿+患儿)*(临床试验+临床研究+临床评估+临床评价+效果评价+干预研究+疗效+随机+盲法+双盲+单盲+对照+三盲+队列+RCT)

E. Wanfang 万方（期刊学位会议）

主题:("舒利迭"+"沙美特罗"*"氟替卡松"+Seretide+Advair+沙美特罗替卡松)*(哮喘)*(儿童+青少年+少年+学龄前+学龄+适龄+幼儿+少儿+患儿)*("临床试验"+"临床研究"+"临床评估"+"临床评价"+"效果评价"+"干预研究"+疗效+随机+盲法+双盲+单盲+对照+三盲+队列+RCT)

F. VIP 维普

R=(舒利迭+沙美特罗*氟替卡松+Seretide+Advair+沙美特罗替卡松)*(哮喘)*(儿童+青少年+少年+学龄前+学龄+适龄+幼儿+少儿+患儿)*(临床试验+临床研究+临床评估+疗效+临床评价+效果评价+疗效观察+干预研究+随机+盲法+双盲+单盲+对照+三盲+队列+RCT) OR M=(舒利迭+沙美特罗*氟替卡松
+Seretide+Advair+沙美特罗替卡松)*(哮喘)*(儿童+青少年+少年+学龄前+学龄+适龄+幼儿+少儿+患儿)*(临床试验+临床研究+临床评估+疗效+临床评价+效果评价+疗效观察+干预研究+随机+盲法+双盲+单盲+对照+三盲+队列+RCT) 466 条，去重后 17 条

G. CBM

#1. 缺省[智能]: 哮喘 OR 主题词:哮喘/全部树/全部副主题词

#2. 缺省[智能]: (舒利迭+沙美特罗*氟替卡松+Seretide+Advair+沙美特罗替卡松)

#3. 缺省[智能]: (儿童+青少年+少年+学龄前+学龄+适龄+幼儿+少儿+患儿)

#4. 缺省[智能]: (临床试验+临床研究+临床评估+临床评价+效果评价+干预研究+疗效+随机+盲法+双盲+单盲+对照+三盲+队列+RCT)

#5. #1 AND #2 AND #3 AND #4

Appendix 2:Characteristics of included studies.

Study ID	Design/Location	Diagnosis	Diagnostic criteria	No. of patients	Age (mean, yrs)	Setting	Length of illness (mean, yrs)	Dosage of SFC	Dosage of MON or MFC	Outcomes	Definition of the Asthma control level assessment and asthma exacerbation
Duan 2016	China	Moderate persistent asthma	Guidelines for the diagnosis and treatment of bronchial asthma in children (revised in 2008) [J]. Chinese medical information guide, 2009,24 (1): 20-21.	105	9.1 (5-13)	hospital (non-specific)	2.9 years	SCF: inhaled 50ug/100ug BID	MFC: oral 5mg QD/125ug BID	Pulmonary function and asthma control level assessment.	Asthma control level assessment: NR
Han 2013	China	Cough variant asthma (CVA)	Guidelines for the diagnosis and treatment of bronchial asthma in children (revised in 2008) [J]. Chinese medical information guide, 2009,24 (1): 20-21.	100	7.66 (4-10)	NR	NR	SCF: inhaled 50ug/100ug BID for 8 weeks	MON: oral 4mg (<4-5 years) or 5mg (≥ 5 years) QD for MON: oral 5mg QD for 8 weeks	Asthma control level assessment.	Asthma control level assessment: NR
Huang 2016	China	Moderate persistent asthma	Guidelines for the diagnosis and treatment of bronchial asthma in children (revised in 2008) [J]. Chinese medical information guide, 2009,24 (1): 20-21.	275	8.2 (NR)	hospital (non-specific)	4.1 years	SCF: inhaled 50ug/100ug BID for 12 weeks	MFC: oral 5mg QD/125ug BID for 12 weeks	Pulmonary function, asthma control level assessment, daytime and nighttime asthma score, days of asthma free symptom and adverse events	Asthma control level assessment: Classification of asthma control level in children according to guidelines for the diagnosis and treatment of bronchial asthma in children (revised in 2008)
Huang 2018	China	Cough variant asthma (CVA)	NR	84	7.53 (6-9)	hospital (non-specific)	NR	SCF: inhaled 50ug/100ug BID for 3 months then QD for next 2 months for 5 months	MON: oral 5mg QD for 5 months	Pulmonary function and asthma control level assessment.	Asthma control level assessment: NR
Jiao 2017	China	Moderate persistent asthma	NR	114	8.7 (4-12)	hospital (non-specific)	NR	SCF: inhaled 25ug/125ug BID for 12 weeks	MFC: oral 5mg QD/one inhalation per time BID for 12 weeks	Asthma control level assessment and adverse events.	Asthma control level assessment: NR

Lemanske 2010	United States	Mild to moderate asthma (uncontrolled)	National Asthma Education and Prevention Program. Expert Panel Report 3 (EPR-3): guidelines for the diagnosis and management of asthma: summary report 2007 [J]. J Allergy Clin Immunol 2007;120: Suppl: S94-S138.	182	11.9 (6-17)	hospital (outpatient)	Mean age at asthma diagnosis: 4 years; Mean age at onset of asthma symptoms: 3.1 years	SCF: inhaled 50ug/100ug BID 16 weeks	MFC: oral 5 or 10mg QD/100ug BID for 16 weeks	NA	-
Liu 2012	China	Mild asthma	Guidelines for the diagnosis and treatment of bronchial asthma in children (revised in 2008) [J]. Chinese medical information guide, 2009,24 (1): 20-21.	33 completer	8.9 (6-14)	hospital (outpatient)	NR	SCF: inhaled 50ug/100ug BID	MON: oral 5mg QD	NR	-
Luo 2013	China	Cough variant asthma (CVA)	NR	120	Median 8.5 (5-11)	hospital (non-specif ic)	NR	SCF: one inhalation per time BID 4 months	MON: oral 5mg QD for 4 months	Asthma control level assessment and recurrence rate.	Asthma control level assessment: NR
Ma 2016	China	Mild to moderate asthma	Guidelines for the diagnosis and treatment of bronchial asthma in children (revised in 2008) [J]. Chinese medical information guide, 2009,24 (1): 20-21.	80	4.4 (NR)	hospital (non-specif ic)	NR	SCF: inhaled 50ug/100ug BID (reduced ¼ dosage for patients' condition and pulmonary functions after 6 and 12 months) for 1year	MON: oral 4mg (<6 years) or 5mg (≥ 6 years) QD for 1 year	Risk of asthma exacerbation, emergency entering and hospitalization and pulmonary function.	Risk of asthma exacerbation: NR
Maspero 2008	Latin America and Turkey	Persistent asthma (uncontrolled , who were not currently receiving anti-inflamma tory treatment)	Standardization of spirometry, 1994 update [J]. Am J Respir Crit Care Med 1995;152:1107-1136.	548	9.3 (6-14)	hospital (outpatient)	<1 year: 7/548; 1-5 years: 198/548; 5-10 years: 297/548; >=10 years: 45/548	SCF: inhaled 50ug/100ug BID 12 weeks	MON: oral 5mg QD for 12 weeks	Risk of asthma exacerbation Adverse events, pulmonary function, symptoms change, and quality of life	Risk of asthma exacerbation: >1 of the following: worsening asthma that required an emergency department visit or hospitalization; an unscheduled doctor visit or contact requiring treatment with oral, parenteral, or inhaled corticosteroids, or treatment with nebulized salbutamol;

Rao 2016	China	Moderate persistent asthma	Guidelines for the diagnosis and treatment of bronchial asthma in children (revised in 2008) [J]. Chinese medical information guide, 2009,24 (1): 20-21.	82	9.2 (5-13)	hospital (non-specific)	2.08 years	SCF: inhaled 50ug/100ug BID for 8 weeks	MFC: oral 4mg QD/125ug BID for 8 weeks	Pulmonary function and asthma control level assessment and C-ACT score.	the use of >12 puffs of salbutamol in a 24-hour period; or the use of >10 puffs of salbutamol on each of 2 consecutive days. Asthma control level assessment: Classification of asthma control level in children according to guidelines for the diagnosis and treatment of bronchial asthma in children (revised in 2008)
Sorkness 2007	United States	Mild to moderate persistent asthma	Standardization of spirometry, 1994 update [J]. Am J Respir Crit Care Med 1995;152:1107-1136.	189	10 (6-14)	community	Mean age at doctor's asthma diagnosis: 4.1 years; Mean age at onset asthma symptoms: 3.1 years	SCF: inhaled 50ug/100ug BID for 48 weeks	MON: oral 5mg QD for 48 weeks	Risk of hospitalization, pulmonary function, score of asthma control questionnaire.	Risk of asthma exacerbation: NR
Wang 2012	China	Cough variant asthma (CVA)	The routine of asthma prevention and treatment in children [J]. Chinese Journal of Pediatrics, 1998,36 (12): 747	95	Median 8.9 (6-12)	hospital (non-specific)	NR	SCF: inhaled 50ug/100ug BID for 12 weeks	MON: oral 5mg QD for 12 weeks	Pulmonary function and asthma control level assessment.	Asthma control level assessment: increase of PEF more than 35% before treatment.
Wang 2014	China	Moderate persistent asthma	Guidelines for the diagnosis and treatment of bronchial asthma in children (revised in 2008) [J]. Chinese medical information guide, 2009,24 (1): 20-21.	60	8.5 (5-14)	hospital (non-specific)	3.22 years	SCF: inhaled 50ug/100ug BID for 12 weeks	MFC: oral 5mg QD/125ug BID for 12 weeks	Pulmonary function, asthma control level assessment, C-ACT score and adverse events.	Asthma control level assessment: Classification of asthma control level in children according to guidelines for the diagnosis and treatment of bronchial asthma in children (revised in 2008)
Wang 2018	China	Moderate persistent asthma	Guidelines for the diagnosis and treatment of bronchial asthma in children (revised in 2008) [J]. Chinese medical information guide, 2009,24 (1): 20-21.	96	9.07 (6-11)	hospital (non-specific)	3.25 years	SCF: inhaled 50ug/100ug BID for 12 weeks	MFC: oral 5mg QD/125ug BID for 12 weeks	Pulmonary function, asthma control level assessment, C-ACT score and adverse events.	Asthma control level assessment: NR

Yuan 2012	China	Cough variant asthma (CVA)	NR	56	7.85 (4-13)	hospital (non-specific)	11.52 weeks	SCF: inhaled 50ug/100ug BID for 8 weeks	MON: oral 4mg QD for 8 weeks	Recurrence rate.	-
Zhang 2009	China	Moderate persistent asthma	Guidelines for the diagnosis and treatment of bronchial asthma in children (revised in 2008) [J]. Chinese medical information guide, 2009,24 (1): 20-21.	75	8.60 (6-13)	hospital (outpatient)	NR	SCF: inhaled 50ug/100ug BID for 12 weeks	MFC: oral 5mg QD/125ug BID for 12 weeks	Pulmonary function, risk of asthma exacerbation and adverse events.	Risk of asthma exacerbation: NR
Zhang 2014	China	Moderate persistent asthma	Guidelines for the diagnosis and treatment of bronchial asthma in children (revised in 2008) [J]. Chinese medical information guide, 2009,24 (1): 20-21.	139	8.01 (5-14)	hospital (outpatient)	2.1 years	SCF: inhaled 25ug/125ug BID for 12 weeks	MFC: oral 5mg QD/125ug BID for 12 weeks	Pulmonary function,asthma control level assessment, daytime and nighttime asthma score, adverse events and the frequency of reliever use.	Asthma control level assessment: Global Initiative for Asthma. Available online at www.ginasthma.com; last accessed Feb. 20, 2007.
Fogel 2010	Europe, Asia, Mexico, and South America	Asthma	NR	154	10.2 (6-14)	NR	NR	SCF: inhaled 100ug/100ug BID	MFC: oral 5mg QD/100ug BID	NA	-
Verini 2007	Italy	mild to moderate persistent asthma	Global Initiative for Asthma. Available online at www.ginasthma.com; last accessed Feb. 20, 2007.	12	9.5 (6-13)	hospital (outpatient)	NR	SCF: inhaled 50ug/100ug BID	MFC: oral 5mg QD/100ug BID	NA	NA
Lenney 2013	United Kingdom	uncontrolled asthma	NR	63 (44 in two treatment groups)	10.39 (6.5-14.67)	NR	NR	SCF: inhaled 50ug/100ug BID for 48 weeks	MFC: oral 5mg QD/100ug BID for 48 weeks	Risk of asthma exacerbation, hospitalization, pulmonary function, quality of life and adverse events.	Risk of asthma exacerbation: NR

NOTE:

SCF = combination salmeterol/fluticasone, MFC = montelukast/fluticasone, MON= Montelukast sodium; BID = twice daily, QD = once daily

C-ACT= Childhood Asthma Control Test

NA = not applicable, NR = not reported

Appendix 3

Table 1:Summary of secondary outcome – change of pulmonary function comparing SFC with MON.

Study	Mean age, year (range)	Diagnosis	Patients	Details of treatment	Timepoints of outcomes	Significant key results
Maspero 2008 (7) Latin America and Turkey	9.3 (6-14)	Persistent asthma	548	SCF: inhaled 50ug/100ug BID vs. MON: oral 5mg QD for 12 weeks	12 weeks	SFC treatment significantly Improved morning PEF (45.82±2.82 L/min vs. 28.7±2.86 L/min, P<0.001), evening PEF (46.2±48.63 L/min vs. 28.0±45.41 L/min, P<0.001) and FEV_1 (0.47±0.017 L vs. 0.3±0.018 L, P<0.001).
Wang 2012 (16) China	8.9 (6-12)	Cough variant asthma	95	SCF: inhaled 50ug/100ug BID vs. MON: oral 5mg QD for 12 weeks	12 weeks	No significant differences in PEF%pred.
Sorkness 2007 (25) U.S.	10 (6-14)	Mild to moderate persistent asthma	189	SCF: inhaled 50ug/100ug BID vs. MON: oral 5mg QD for 48 weeks	48 weeks	SFC treatment significantly improved morning PEF%pred (mean difference 4.68, 95%CI: 1.88,7.48), evening PEF%pred (mean difference 4.88, 95%CI: 2.23,7.52) and FEV_1%pred (mean difference 4.20, 95%CI: 1.02,7.39).
Huang 2018 (11) China	7.53 (6-9)	Cough variant asthma	84	SCF: inhaled 50ug/100ug BID vs. MFC: oral 5mg QD/100ug BID for 48 weeks	5 months	MON treatment significantly improved FEV_1%pred (66.16±6.05 vs. 78.09±7.46, p<0.05).
Ma 2016 (6) China	4.4 (NR)	Mild to moderate asthma	80	SCF: inhaled 50ug/100ug BID (reduced ¼ dosage for patients' condition and pulmonary functions after 6 and 12 months) vs. MON: oral 4mg (<6 years) or 5mg (≥ 6 years) QD for 1 year	6months, 12months and 18 months	No significant differences in PEF%pred and FEV_1%pred.

NOTE: SCF = combination salmeterol/fluticasone, MFC = montelukast/fluticasone, MON= Montelukast sodium; BID = twice daily, QD = once daily; NR = not reported; PEF = peak expiratory flow; FEV_1 = forced expiratory flow rate in one second; PEF%pred= percentage of predicted PEF; FEV_1%pred = percentage of predicted FEV_1.

Table 2:Summary of secondary outcome – change of pulmonary function could not be performed Meta analysis Comparing SFC with MFC.

Study	Mean age, year (range)	Diagnosis	Patients	Details of treatment	Timepoints of outcomes	Significant key results
Rao 2016 (12) China	9.2 (5-13)	Moderate persistent asthma	82	SCF: inhaled 50ug/100ug BID vs. MFC: oral 4mg QD/125ug BID for 8 weeks	5 weeks	No significant differences in FEV_1%pred, PEF%pred or $FEF_{25\%-75\%}$.
Wang 2018 (17) China	9.07 (6-11)	Moderate persistent asthma	96	SCF: inhaled 50ug/100ug BID vs. MFC: oral 5mg QD/125ug BID for 12 weeks	4 weeks and 12 weeks	SFC treatment significantly Improved FEV_1 at 4 weeks (78.52±7.03L vs. 70.67±6.58L, p<0.01) and 12 weeks (89.61±9.49 vs. 74.15±10.33L, p<0.01), respectively.
Lenney 2013 (29) U.K.	10.39 (6.5-14.67)	Uncontrolled asthma	63	SCF: inhaled 50ug/100ug BID vs. MFC: oral 5mg QD/100ug BID for 48 weeks	24 weeks and 48 weeks	No significant differences in FEV_1%pred.
Duan 2016 (20) China	9.1 (5-13)	Moderate persistent asthma	105	SCF: inhaled 50ug/100ug BID vs. MFC: oral 5mg QD/125ug BID	NR	SFC treatment significantly improved FEV_1 (85.9±14.7L vs. 73.6±12.3L, p<0.05) and PFE%pred (87.6±15.3L vs. 72.0±12.9L, p<0.05) after treatment.

NOTE: SCF = combination salmeterol/fluticasone, MFC = montelukast/fluticasone, MON= Montelukast sodium; BID = twice daily, QD = once daily; NR = not reported; FEV_1 = forced expiratory flow rate in one second; PEF%pred= percentage of predicted peak expiratory flow; FEV_1%pred = percentage of predicted FEV_1; $FEF_{25\%-75\%}$ = Forced expiratory flow at 25-75% of forced vital capacity.

Appendix 4: Meta-analysis forest plots of Pulmonary function comparing SFC with MFC.

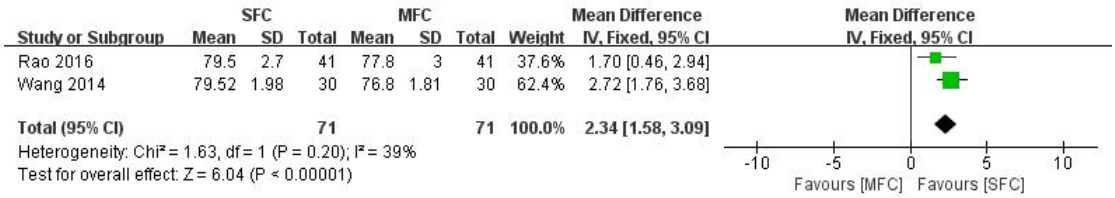


Figure 1: The results of FEV_1%pred after 4 weeks treatment.

NOTE: FEV_1%pred = percentage of predicted forced expiratory flow rate in one second.

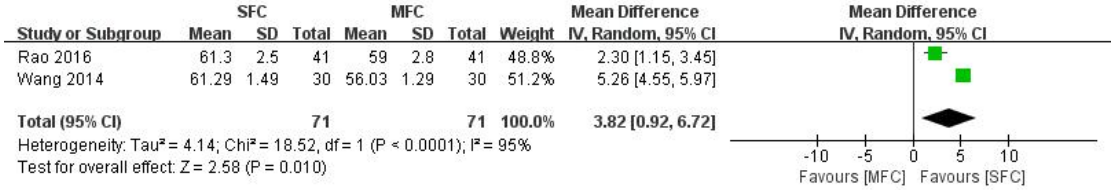


Figure 2:The results of $FEF_{25\%}$ after 4 weeks treatment.

NOTE: $FEF_{25\%}$ = Forced expiratory flow at 25% of forced vital capacity.

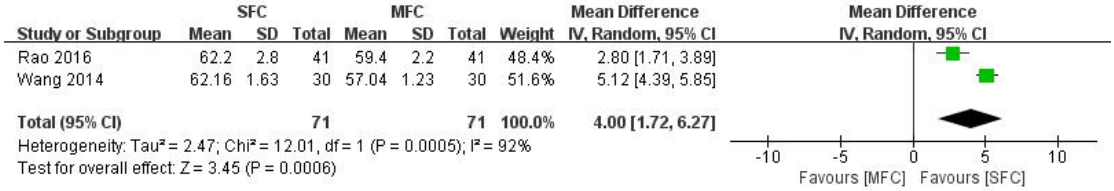


Figure 3:The results of $FEF_{50\%}$ after 4 weeks treatment.

NOTE: $FEF_{50\%}$ = Forced expiratory flow at 50% of forced vital capacity.

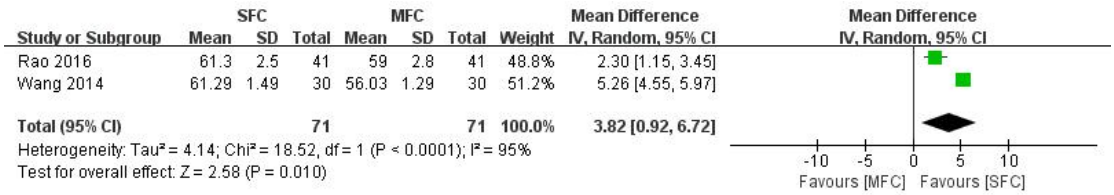


Figure 4:The results of $FEF_{75\%}$ after 4 weeks treatment.

NOTE: $FEF_{75\%}$ = Forced expiratory flow at 75% of forced vital capacity.

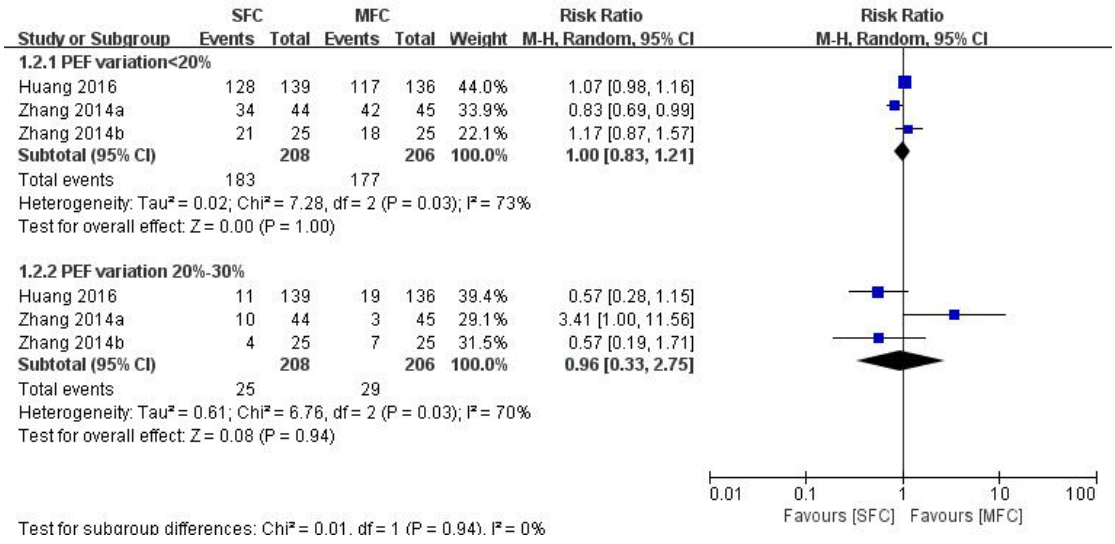


Figure 5:The results of PEF variation after 12 weeks treatment.

NOTE: PEF = peak expiratory flow.

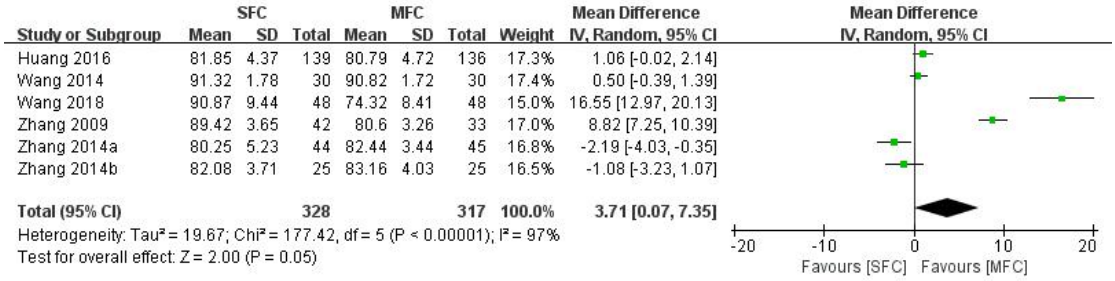


Figure 6:The results of PEF_1%pred after 12 weeks treatment.

NOTE: PEF = peak expiratory flow.

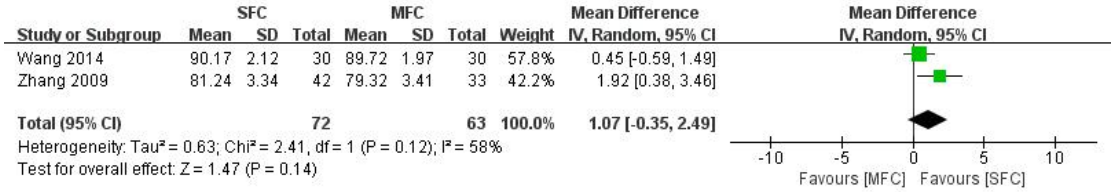


Figure 7:The results of FEV_1%pred after 12 weeks treatment.

NOTE: FEV_1%pred = percentage of predicted forced expiratory flow rate in one second.

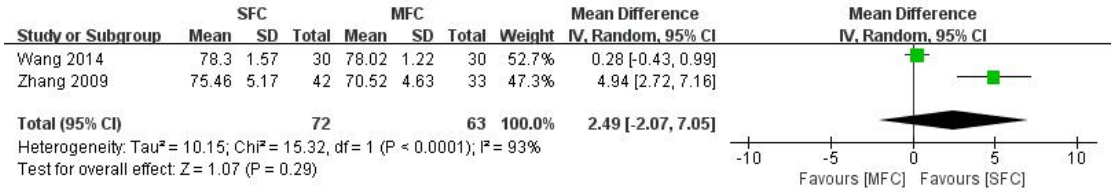


Figure 8:The results of $FEF_{25\%}$ after 12 weeks treatment.

NOTE: $FEF_{25\%}$ = Forced expiratory flow at 25% of forced vital capacity.

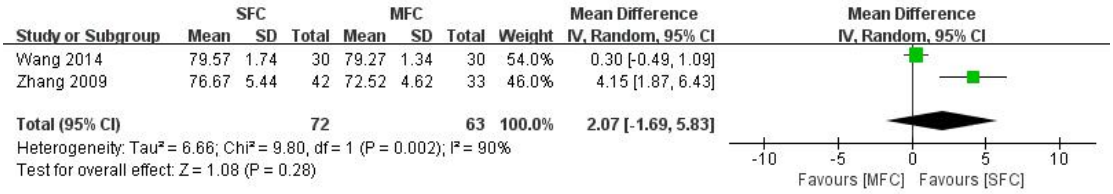


Figure 9:The results of $FEF_{50\%}$ after 12 weeks treatment.

NOTE: $FEF_{50\%}$ = Forced expiratory flow at 50% of forced vital capacity.

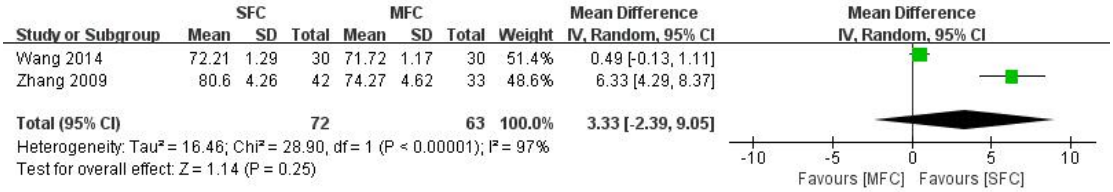


Figure 10:The results of $FEF_{75\%}$ after 12 weeks treatment.

NOTE: $FEF_{75\%}$ = Forced expiratory flow at 75% of forced vital capacity.

Appendix 5

Table 1:Summary of secondary outcome –asthma control level comparing SFC with MON.

Study	Mean age, year (range)	Diagnosis	Patients	Details of treatment	Timepoints of outcomes	Significant key results
Yuan 2012 (14) China	7.85 (4-13)	Cough variant asthma	56	SCF: inhaled 50ug/100ug BID vs. MON: oral 4mg QD for 8 weeks	8 weeks	No significant differences in recurrence rate.
Han 2013 (13) China	7.66 (4-10)	Cough variant asthma	100	SCF: inhaled 50ug/100ug BID vs. MON: oral 4mg (<4-5 years) or 5mg (≥ 5 years) QD for MON: oral 5mg QD for 8 weeks	8 weeks	SFC treatment improved clinical effective rate significantly (94% vs. 84%, P<0.05).
Maspero 2008 (7) Latin, America and Turkey	9.3 (6-14)	Persistent asthma	548	SCF: inhaled 50ug/100ug BID vs. MON: oral 5mg QD for 12 weeks	12 weeks	SFC group had more asthma symptom-free days (OR, 1.74; 95% CI, 1.07-2.82) and more rescue-free days (OR, 3.24; 95% CI, 2.09-5.02).
Wang 2012 (16) China	8.9 (6-12)	Cough variant asthma	95	SCF: inhaled 50ug/100ug BID vs. MON: oral 5mg QD for 12 weeks	12 weeks	No significant differences in proportion of patients in complete control.
Luo 2013 (15) China	8.5 (5-11)	Cough variant asthma	120	SCF: one inhalation per time BID vs. MON: oral 5mg QD for 4 months	24 weeks and 48 weeks	No significant differences in recurrence rate.
Sorkness 2007 (25) U.S.	10 (6-14)	Mild to moderate persistent asthma	189	SCF: inhaled 50ug/100ug BID vs. MON: oral 5mg QD for 48 weeks	48 weeks	SFC group had higher proportion of asthma symptom-free days (mean difference 10.9%, 95%CI, 2.6-19.3). No significant differences in ACQ score.
Huang 2018 (11) China	7.53 (6-9)	Cough variant asthma	84	SCF: inhaled 50ug/100ug BID for 3 months then QD for next 2 months vs. MON: oral 5mg QD for 5 months	5 months	SFC treatment improved clinical effective rate significantly (81% vs. 95.2%, p<0.05).

NOTE: SCF = combination salmeterol/fluticasone, MFC = montelukast/fluticasone, MON= Montelukast sodium; BID = twice daily, QD = once daily; NR = not reported.

Table 2:Summary of secondary outcome –asthma control level which could not be performed Meta analysis Comparing SFC with MFC.

Study	Mean age, year (range)	Diagnosis	Patients	Details of treatment	Timepoints of outcomes	Significant key results
Rao 2016 (12) China	9.2 (5-13)	Moderate persistent asthma	82	SCF: inhaled 50ug/100ug BID vs. MFC: oral 4mg QD/125ug BID for 8 weeks	8 weeks	No significant difference in clinical effective rate or C-ACT score.
Wang 2018 (17) China	9.07 (6-11)	Moderate persistent asthma	96	SCF: inhaled 50ug/100ug BID vs. MFC: oral 5mg QD/125ug BID for 12 weeks	12 weeks	SFC treatment improved clinical effective rate significantly (87.5% vs. 72.9%, p=0.02).
Jiao 2017 (19) China	8.7 (4-12)	Moderate persistent asthma	114	SCF: inhaled 25ug/125ug BID vs. MFC: oral 5mg QD/one inhalation per time BID for 12 weeks	12 weeks	MFC treatment decreased clinical effective rate significantly (68.4% vs. 96.5%, p=0.001).
Duan 2016 (20) China	9.1 (5-13)	Moderate persistent asthma	105	SCF: inhaled 50ug/100ug BID vs. MFC: oral 5mg QD/125ug BID	NR	SFC treatment improved clinical effective rate significantly (86.5% vs. 71.7%, p<0.05)

NOTE: SCF = combination salmeterol/fluticasone, MFC = montelukast/fluticasone, MON= Montelukast sodium; BID = twice daily, QD = once daily; NR = not reported; C-ACT= Childhood Asthma Control Test.

Appendix 6:Meta-analysis forest plots of asthma control level comparing SFC with MFC.

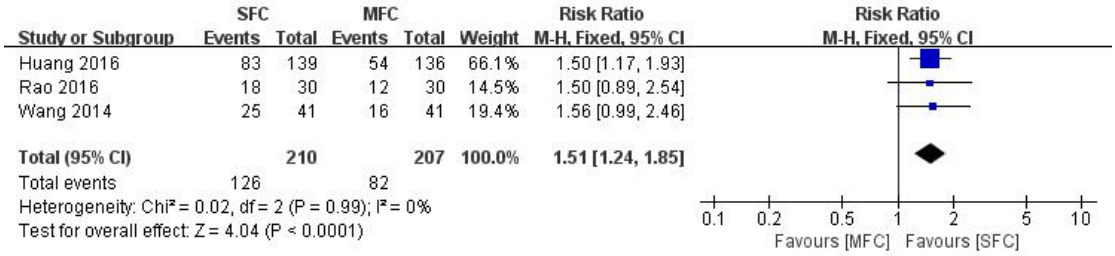


Figure 1:The results of full controlled after 4 weeks treatment.

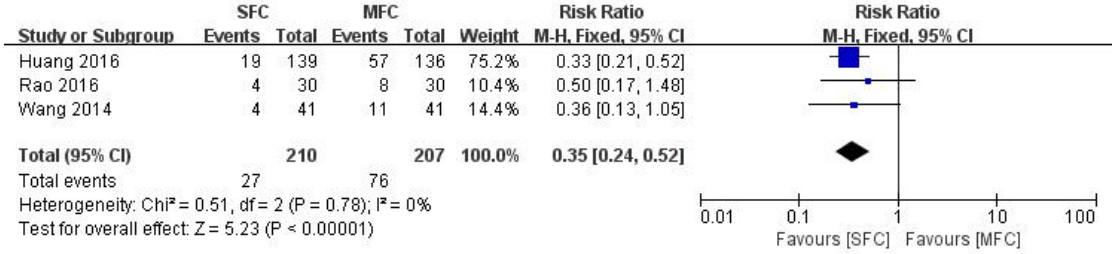


Figure 2:The results of uncontrolled after 4 weeks treatment.

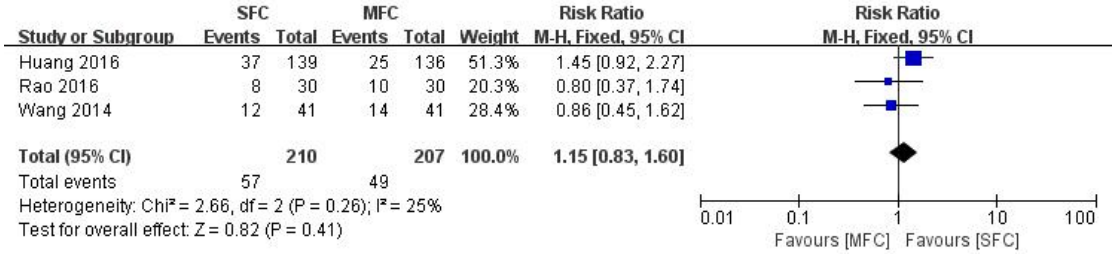


Figure 3:The results of partially controlled after 4 weeks treatment.

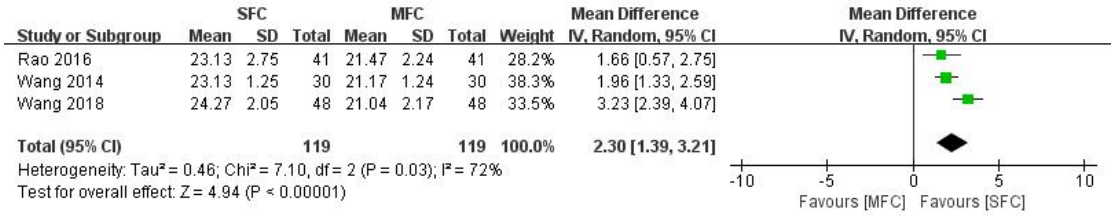


Figure 4:The results of C-ACT score after 4 weeks treatment.

NOTE: C-ACT= Childhood Asthma Control Test.

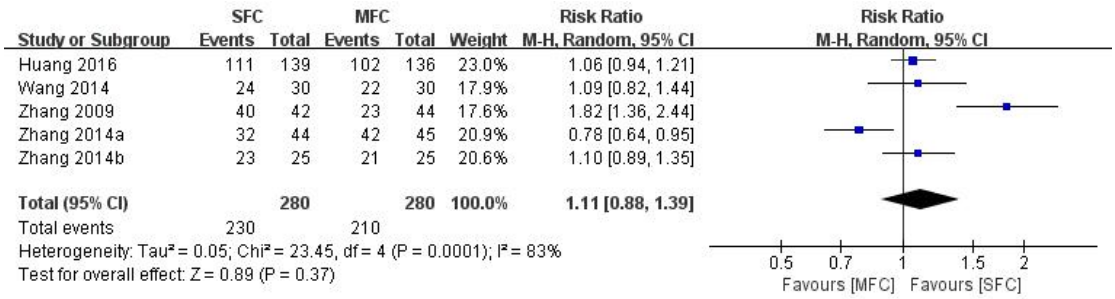


Figure 5:The results of full controlled after 12 weeks treatment.

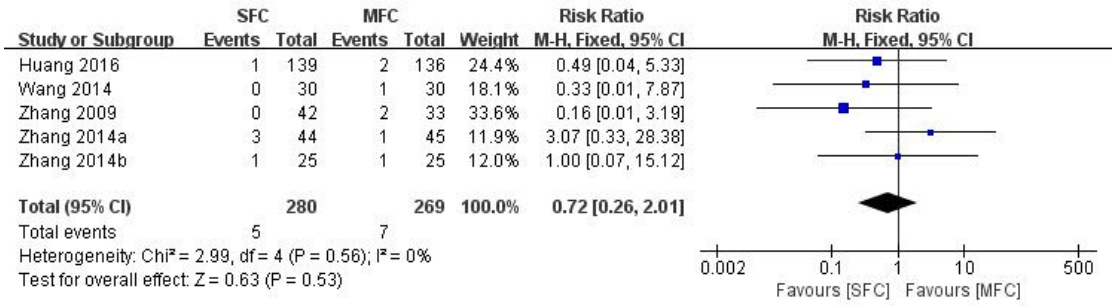


Figure 6:The results of uncontrolled after 12 weeks treatment.

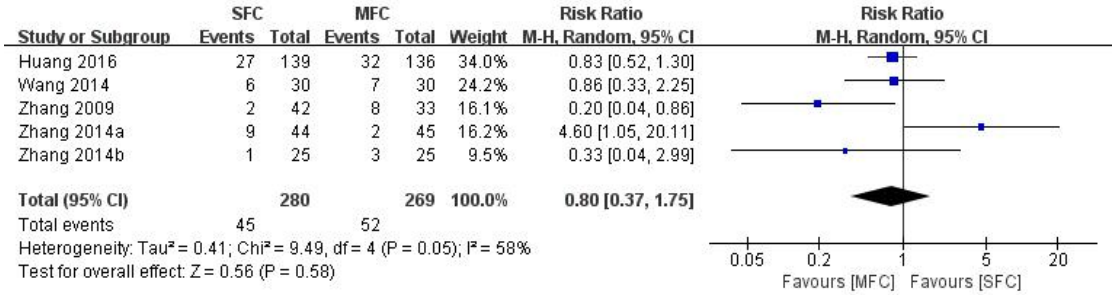


Figure 7:The results of partially controlled after 12 weeks treatment.

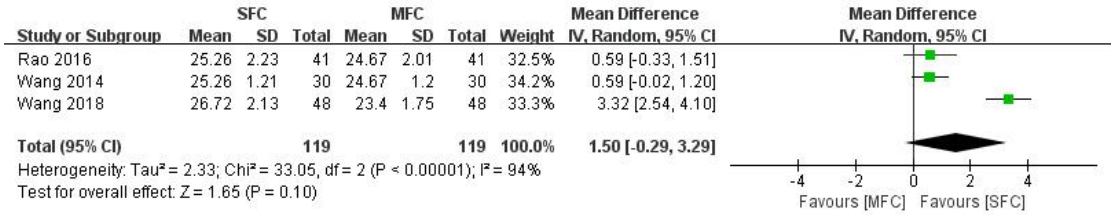


Figure 8:The results of C-ACT score after 12 weeks treatment.

NOTE: C-ACT= Childhood Asthma Control Test.

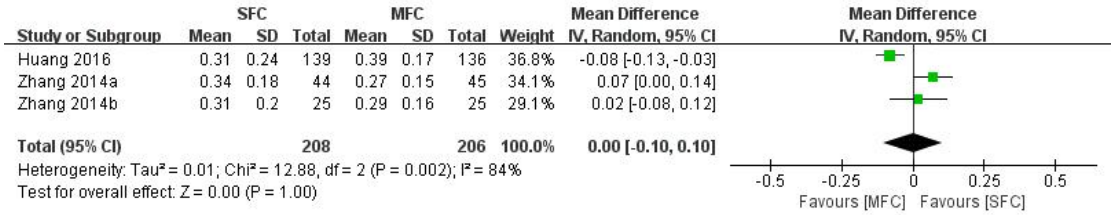


Figure 9:The results of day-time asthma score after 12 weeks treatment.

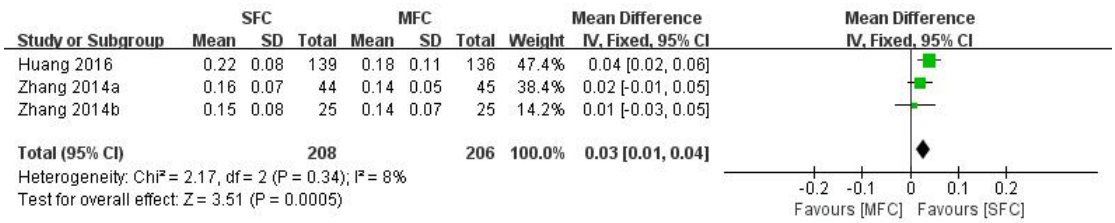


Figure 10:The results of night-time asthma score after 12 weeks treatment.

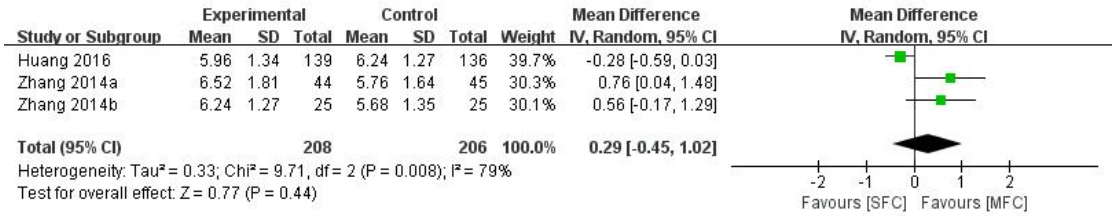


Figure 11:The results of frequency of reliever use after 12 weeks treatment.

Appendix 7:Summary of secondary outcome –quality of life comparing SFC with MON and SFC with MFC.

Study	Mean age, year (range)	Diagnosis	Patients	Intervention & Comparison	Details of treatment	Timepoints of outcomes	Significant key results
Maspero 2008 (7) Latin America and Turkey	9.3 (6-14)	Persistent asthma	548	SFC vs. MON	SCF: inhaled 50ug/100ug BID vs. MON: oral 5mg QD for 12 weeks	12 weeks	No significant differences in PAQLQ.
Lenney 2013 (29) U.K.	10.39 (6.5-14.67)	Uncontrolled asthma	63	SFC vs. MFC	SCF: inhaled 50ug/100ug BID vs. MFC: oral 5mg QD/100ug BID for 48 weeks	24 weeks and 48 weeks	SFC treatment significantly improved the change of PACQLQ score (1.5±0.17 vs. 1.0±0.17; P = 0.028). No significant differences in PAQLQ score.

NOTE: SCF = combination salmeterol/fluticasone, MFC = montelukast/fluticasone, MON= Montelukast sodium; BID = twice daily, QD = once daily; PAQLQ = Pediatric Asthma Quality of Life Questionnaire; PACQLQ = Pediatric Asthma Caregiver's Quality of Life Questionnaire.

Appendix 8

Table 1:Summary of secondary outcome –adverse events comparing SFC with MON.

Study	Mean age, year (range)	Diagnosis	Patients	Details of treatment	Timepoints of outcomes	Significant key results
Maspero 2008 (7) Latin, America and Turkey	9.3 (6-14)	Persistent asthma	548	SCF: inhaled 50ug/100ug BID vs. MON: oral 5mg QD for 12 weeks	12 weeks	No significant differences in overall adverse events
Ma 2016 (6) China	4.4 (NR)	Mild to moderate asthma	80	SCF: inhaled 50ug/100ug BID (reduced ¼ dosage for patients' condition and pulmonary functions after 6 and 12 months) vs. MON: oral 4mg (<6 years) or 5mg (≥ 6 years) QD for 1 year	6months, 12months and 18 months	No significant differences in overall adverse events.

NOTE: SCF = combination salmeterol/fluticasone, MON= Montelukast sodium; BID = twice daily, QD = once daily; NR = not reported.

Table 2: Summary of secondary outcome – adverse events which could not be performed Meta analysis comparing SFC with MFC.

Study	Mean age, year (range)	Diagnosis	Patients	Details of treatment	Timepoints of outcomes	Significant key results
Jiao 2017 (19) China	8.7 (4-12)	Moderate persistent asthma	114	SCF: inhaled 25ug/125ug BID vs. MFC: oral 5mg QD/one inhalation per time BID for 12 weeks	12 weeks	No significant difference in risk of overall adverse events.
Wang 2014 (18) China	8.5 (5-14)	Moderate persistent asthma	60	SCF: inhaled 50ug/100ug BID vs. MFC: oral 5mg QD/125ug BID for 12 weeks	NR	No significant difference in risk of adverse events.
Wang 2018 (17) China	9.07 (6-11)	Moderate persistent asthma	96	SCF: inhaled 50ug/100ug BID vs. MFC: oral 5mg QD/125ug BID for 12 weeks	NR	No significant difference in risk of adverse events.
Zhang 2009 (21) China	8.6 (6-13)	Moderate persistent asthma	75	SCF: inhaled 50ug/100ug BID vs. MFC: oral 5mg QD/125ug BID for 12 weeks	NR	No significant difference in risk of adverse events.

NOTE: SCF = combination salmeterol/fluticasone, MFC = montelukast/fluticasone, BID = twice daily, QD = once daily; NR = not reported.

Appendix 9:Meta-analysis forest plots of adverse events after 12 weeks treatment comparing SFC with MFC.

